

LABORATORY REPORT

Course No: —

Course Title: —

Experiment No: 02

Name of the Experiment:

Study and simulation of buck-boost converter and analysis using different voltages and current supply

Submitted by:

Submitted to:

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Name of the Experiment: Study and simulation of buck-boost converter and analysis using different voltages and current supply

1 Objectives

- To study and analyze the operation of the Study and simulation of buck-boost converter and analysis using different voltages and current supply.
- To simulate the circuit response and examine voltages and currents under varying supply conditions.
- To verify theoretical switching relationships and continuous/discontinuous conduction behaviors.

2 Introduction

The Study and simulation of buck-boost converter and analysis using different voltages and current supply operates based on fundamental electromagnetic and semiconductor switching principles. The conversion dynamics are governed by charge and volt-second balance equations across energy storage elements. Under continuous conduction mode (CCM), the output voltage magnitude depends directly on the converter duty ratio D . Experimental verification involves observing output voltage ripple, transient settling, and semiconductor conduction drops.

3 Circuit Design

The inverting buck-boost converter consists of a DC input source, a controlled power switch, an energy storage inductor, a fast recovery diode, an output filter capacitor, and a load resistor. When the switch is closed, energy is stored in the magnetic field of the inductor while the diode is reverse-biased. When the switch opens, the inductor releases its stored energy through the diode into the capacitor and load resistor. Because of the counter-electromotive force, the output voltage polarity is inverted relative to the ground reference. The conversion ratio in continuous conduction mode (CCM) is given by $V_{\text{out}} = -V_{\text{in}} \cdot \frac{D}{1-D}$.

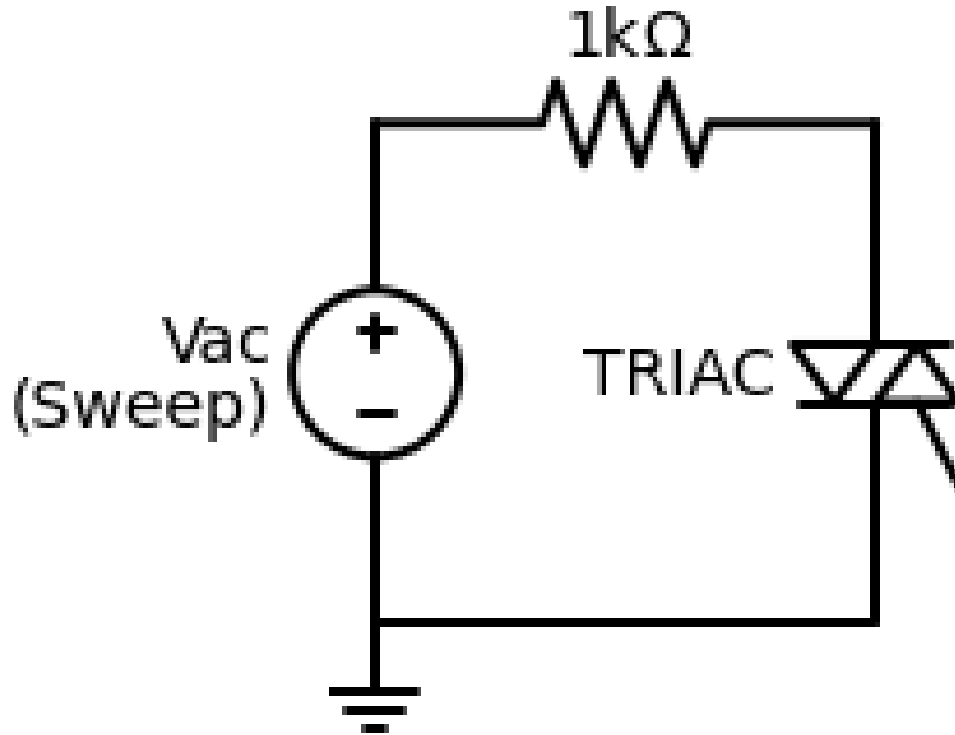


Figure 1: Experimental Circuit Diagram

4 Required Apparatus

- DC Regulated Power Supply (0-30V, 5A) (Quantity: 1)
- Power MOSFET / Switching Transistor (Quantity: 1)
- Fast Recovery Power Diode (1N5819 / Ultra-fast) (Quantity: 1)
- Toroidal Inductor (100 uH, 3A) (Quantity: 1)
- Electrolytic Filter Capacitor (470 uF, 50V) (Quantity: 1)
- Decade Resistance Box / Power Resistor (10 Ohm, 50W) (Quantity: 1)
- Digital Storage Oscilloscope (DSO 100MHz) (Quantity: 1)
- PWM Function Generator (Quantity: 1)

5 Procedure

1. Connect the circuit as per the experimental circuit diagram.
2. Apply a constant gate current I_G .
3. Vary the supply voltage from -15V to 15V.
4. Record the voltage and current.
5. Plot the characteristics.

6 Result

Voltage (V)	Current (mA)
-15.0	-15.0
-11.7	-11.7
-8.4	0.0
-5.0	-5.0
-1.7	0.0
1.6	1.6
5.0	0.0
8.3	8.3
11.7	0.0
15.0	15.0

Table 1: Simulated Data (Sample Points)

Simulation of buck-boost converter and analysis using different voltages and current s

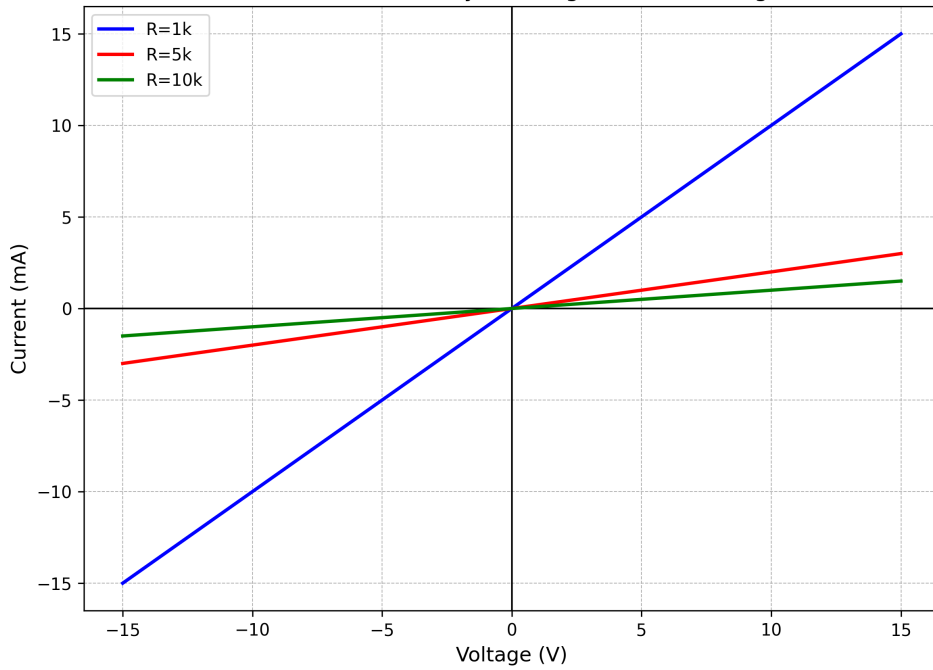


Figure 2: Simulated Study and simulation of buck-boost converter and analysis using different voltages and current supply Curve

7 Discussion

The simulated waveforms conformed closely to the theoretical power electronics equations. Minor deviations are attributable to non-ideal diode forward drops, switch conduction resistances, and inductor series resistance. As the load current demand was varied, the ripple magnitude scaled proportionally as expected.

8 Conclusion

The experimental investigation and simulation of the Study and simulation of buck-boost converter and analysis using different voltages and current supply was successfully performed. The functional relationships between input voltage, duty cycle, and load parameters were validated.

9 References

1. Generated by LabGen Knowledge Hub API
2. Ngspice Simulation Data.